Priprema za prvi kolokvijum

Matematička analiza 2

30. 11. 2022.

Prvi kolokvijum 2020.

Zadatak 1. U zavisnosti od realnog parametra $\alpha \geq 0$, ispitati apsolutnu i uslovnu konvergenciju reda $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^{\alpha} \cdot \sqrt{n^2+1}}$.

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} \frac{1}{n^{\alpha} \cdot \sqrt{n^2+1}}$$

$$\frac{1}{n^{\alpha} \cdot \sqrt{n^2+1}} \sim \frac{1}{n^{\alpha} \cdot n} = \frac{1}{n^{\alpha+1}}$$

$\alpha+1>1$ konv.

tj. $\boxed{\alpha > 0}$

za $\alpha>0$ $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^{\alpha} \cdot \sqrt{n^2+1}}$ konv. aps.

za $\alpha = 0$: $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n^2+1}}$ aps. div

Lajb. krit:

1. $\lim_{n\to\infty} |a_n| = \lim_{n\to\infty} \frac{1}{\sqrt{n^2+1}} = 0$ ✓

2. $|a_{n+1}| \leq |a_n|$

$$\frac{1}{\sqrt{(n+1)^2+1}} \leq \frac{1}{\sqrt{n^2+1}}$$ ✓

$\Rightarrow$ red obično konv.

red uslovno konv.

Zadatak 2. Funkciju $f(x) = (x+1)e^{2x} + \ln\sqrt{2x^2+7x+6}$ razviti u Tejlorov red u okolini tačke $x_0 = -1$ i odrediti oblast konvergencije dobijenog razvoja.

$2x^2 + 7x + 6 = 0$

$x_{12} = \dfrac{-7 \pm \sqrt{49-48}}{4} = \dfrac{-7 \pm 1}{4}$ ⟨ $-\frac{3}{2}$, -2

$2\left(x+\frac{3}{2}\right)(x+2) = (2x+3)(x+2)$

$f(x) = (x+1)\cdot e^{2x} + \frac{1}{2}\left(\ln(2x+3) + \ln(x+2)\right) =$

$= (x+1)\cdot e^{2(x+1)-2} + \frac{1}{2}\left(\ln\left(2(x+1)+1\right) + \ln\left((x+1)+1\right)\right) =$

$= (x+1)\cdot e^{-2}\cdot \sum_{n=0}^{\infty} \frac{2^n\cdot(x+1)^n}{n!} + \frac{1}{2}\left(\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}\cdot 2^n\cdot(x+1)^n + \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}(x+1)^n\right)$

$= \frac{1}{e^2}\cdot\sum_{n=0}^{\infty}\frac{2^n\cdot(x+1)^{n+1}}{n!} + \frac{1}{2}\cdot\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}\left(2^n+1\right)\cdot(x+1)^n =$

$$= \frac{1}{e^2} \cdot \sum_{n=1}^{\infty} \frac{2^{n-1} \cdot (x+1)^n}{(n-1)!} + \frac{1}{2} \cdot \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (2^n+1) \cdot (x+1)^n =$$

$$= \sum_{n=1}^{\infty} \left(\frac{2^{n-1}}{e^2 \cdot (n-1)!} + \frac{1}{2} \cdot \frac{(-1)^{n-1}}{n} (2^n+1) \right) \cdot (x+1)^n$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \; x \in \mathbb{R}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \cdot x^n, \quad -1 < x \leq 1$$

$2x^2 + 7x + 6 > 0,$

$2x + 3 > 0,$

$x + 2 > 0,$

$-1 < 2(x+1) \leq 1$, $\quad -1 < x+1 \leq 1$

$-\frac{1}{2} < x+1 \leq \frac{1}{2}$ $\quad -2 < x \leq 0$

$-\frac{3}{2} < x \leq -\frac{1}{2}$

$\longrightarrow$ oblast konv : $x \in \left(-\frac{3}{2}, -\frac{1}{2}\right]$

Prvi kolokvijum 2020.

Zadatak 3. Odrediti oblast konvergencije i naći sumu reda

$$\sum_{n=0}^{\infty} \frac{n^2+n-2}{n+1}(x-2020)^n.$$

smena: $t = x - 2020 \Rightarrow \sum_{n=0}^{\infty} \frac{n^2+n-2}{n+1} \cdot t^n$

$$R = \frac{1}{\lim_{n\to\infty} \sqrt[n]{\frac{n^2+n-2}{n+1}}} = \frac{1}{\lim_{n\to\infty} \frac{\sqrt[n]{n^2+n-2} \to 1}{\sqrt[n]{n+1} \to 1}} = 1 \Longrightarrow \begin{array}{l} -1 < x - 2020 < 1 \\ 2019 < x < 2021 \end{array}$$

$x = 2021$: $\sum_{n=0}^{\infty} \frac{n^2+n^2-2}{n+1}$ $\left(\lim_{n\to\infty} \frac{n^2+n-2}{n+1} \neq 0 \right)$ div.

$x = 2019$: $\sum_{n=0}^{\infty} \frac{n^2+n-2}{n+1} (-1)^n$, aps. div., obično div. (nisu isp. uslovi Lajb. krit.)

$\longrightarrow$ oblast konv: $x \in (2019, 2021)$

$$\frac{n^2+n-2}{n+1} = \frac{n(n+1)-2}{n+1} = n - \frac{2}{n+1}$$

$$\sum_{n=0}^{\infty} \frac{n^2+n-2}{n+1} t^n = \sum_{n=0}^{\infty} \left(n - \frac{2}{n+1}\right) \cdot t^n = \sum_{n=0}^{\infty} n t^n - 2 \cdot \sum_{n=0}^{\infty} \frac{t^n}{n+1} = \circledast$$

$$\sum_{n=0}^{\infty} n t^n = t \cdot \sum_{n=0}^{\infty} n \cdot t^{n-1} = t \cdot \sum_{n=0}^{\infty} (t^n)' = t \cdot \left(\sum_{n=0}^{\infty} t^n\right)' = t \cdot \left(\frac{1}{1-t}\right)' = t \cdot \frac{1}{(1-t)^2}$$

$$\sum_{n=0}^{\infty} \frac{t^n}{n+1} \overset{*}{=} \frac{1}{t} \cdot \sum_{n=0}^{\infty} \frac{t^{n+1}}{n+1} = \frac{1}{t} \cdot \sum_{n=0}^{\infty} \int_0^t s^n \, ds = \frac{1}{t} \int_0^t \sum_{n=0}^{\infty} s^n \, ds = \frac{1}{t} \cdot \int_0^t \frac{1}{1-s} \, ds =$$

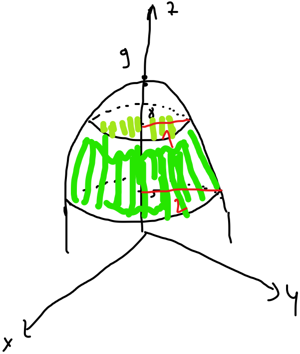
$$= \frac{1}{t} \cdot \left(-\ln|1-s|\right)\Big|_0^t = -\frac{1}{t}\left(\ln|1-t| - \underbrace{\ln|1-0|}_{0}\right) = -\frac{\ln|1-t|}{t}$$

$$\circledast = \frac{t}{(1-t)^2} + \frac{2\ln|1-t|}{t}, \quad t = x - 2020$$

$t = 0: \quad s = -2$

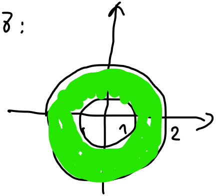
Prvi kolokvijum 2020.

Zadatak 4. Izračunati površinu dela paraboloida $z = 9 - x^2 - y^2$ između ravni $z = 5$ i $z = 8$.



presek: $9 - x^2 - y^2 = 8$, $x^2 + y^2 = 1$ | $9 - x^2 - y^2 = 5$, $x^2 + y^2 = 4$

D:



$x = \rho \cdot \cos\varphi,\ y = \rho \cdot \sin\varphi$

$\rho \in [1, 2],\ \varphi \in [0, 2\pi]$

$z = 9 - x^2 - y^2$

$z'_x = -2x \qquad z'_y = -2y$

$$S = \iint_{D} \sqrt{1 + (z'_x)^2 + (z'_y)^2}\, dx\,dy = \iint_{D} \sqrt{1 + 4x^2 + 4y^2}\, dx\,dy =$$

$$= \int_0^{2\pi}\int_1^{2} \sqrt{1 + 4\rho^2} \cdot \rho\, d\rho\, d\varphi = \int_0^{2\pi} \frac{1}{12} \cdot (1 + 4\rho^2)^{3/2} \Big|_1^2 d\varphi =$$

$$= \frac{1}{12} \cdot \int_0^{2\pi} \left(17^{3/2} - 5^{3/2}\right) d\varphi = \frac{17\sqrt{17} - 5\sqrt{5}}{12} \cdot \varphi \Big|_0^{2\pi} = \boxed{\frac{17\sqrt{17} - 5\sqrt{5}}{6} \cdot \pi}$$

$$\int \sqrt{1 + 4\rho^2}\, \rho\, d\rho = \left(\begin{array}{l} t = 1 + 4\rho^2 \\ dt = 8\rho\, d\rho \end{array} \right) = \frac{1}{8} \int \sqrt{t}\, dt = \frac{1}{8} \cdot \frac{t^{3/2}}{3/2} + C = \frac{1}{12} \cdot (1 + 4\rho^2)^{3/2} + C$$

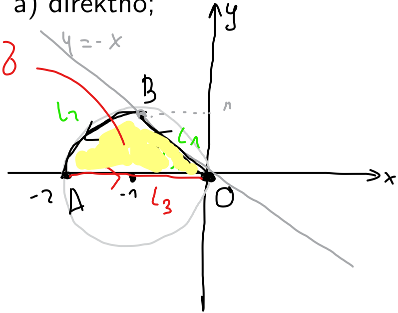
Zadatak 5. Izračunati vrednost krivolinijskog integrala $\int_L 2y\,dx$ ako je L

$(x+1)^2+y^2=1$

$\{(x,y) : y=-x, -1 \le x \le 0\} \cup \{(x,y) : x^2+y^2=-2x,\ x \le -1, 0 \le y \le 1\}$

orijentisana od koordinatnog početka ka tački $A(-2,0)$:

a) direktno;



b) primenom Grinove formule.

$L = L_1 \cup L_2$

$\int_L = \int_{L_1} + \int_{L_2} = -1 - \frac{\pi}{2}$

$L_1: \quad y = -x$

param: $x = t,\ y = -t,\ t \in [-1, 0]$

$x' = 1,\ y' = -1$

$$\int_{L_1} 2(-t)\cdot 1\, dt = -2\int_0^{-1} t\, dt = -\cancel{2}\cdot\frac{t^2}{\cancel{2}}\Big|_0^{-1} = -(1-0) = -1$$

$\sin^2 t = \dfrac{1-\cos 2t}{2}$

$L_2: \quad (x+1)^2 + y^2 = 1$

param: $x = -1 + \cos t,\ y = \sin t,\ t \in [\frac{\pi}{2}, \pi]$

$x' = -\sin t,\ y' = \cos t$

$$\int_{L_2} 2\sin t\cdot(-\sin t)\, dt = -2\int_{\frac{\pi}{2}}^{\pi} \sin^2 t\, dt = -\cancel{2}\int_{\frac{\pi}{2}}^{\pi} \frac{1-\cos 2t}{\cancel{2}}\, dt =$$

$$= -\int_{\frac{\pi}{2}}^{\pi} dt + \int_{\frac{\pi}{2}}^{\pi} \cos 2t\, dt = -t\Big|_{\frac{\pi}{2}}^{\pi} + \frac{1}{2}\cdot\sin 2t\Big|_{\frac{\pi}{2}}^{\pi} =$$

$$= -\left(\pi - \frac{\pi}{2}\right) + \frac{1}{2}\left(\underbrace{\sin 2\pi}_{0} - \underbrace{\sin \pi}_{0}\right) = -\frac{\pi}{2}$$

b.) $L^* = L \cup L_3$ - zat. poz. orijentisana kriva

$$\int_{L^*} = \iint_{\delta} (Q_x - P_y)\,dx\,dy \Longleftrightarrow \int_L + \int_{L_3} = \iint_{\delta} \Longrightarrow \int_L = \iint_{\delta} - \int_{L_3}$$

L_3 : $y = 0$

param: $x = t$, $y = 0$, $t \in [-2, 0]$, $x' = 1$, $y' = 0$

$$\int_{L_3} = \int_{-2}^{0} \underbrace{2 \cdot 0 \cdot 1}_{0}\,dt = 0$$

$$P = 2y \qquad Q = 0$$

$$P_y = 2 \qquad Q_x = 0$$

$$\iint_{\mathcal{D}} (Q_x - P_y)\, dx\, dy = \iint_{\mathcal{D}} (0-2)\, dx\, dy = -2 \iint_{\mathcal{D}} dx\, dy = -2\, P(\mathcal{D}) =$$

$$= -2\left(\frac{1}{4}\cdot 1^2 \cdot \pi + \frac{1\cdot 1}{2}\right) = -\frac{\pi}{2} - 1$$